Relationships: Scatter, Bubble & Hexbin

Workshop 3 — Module 4

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Data Visualization for Data Scientists
UNYT Tirana

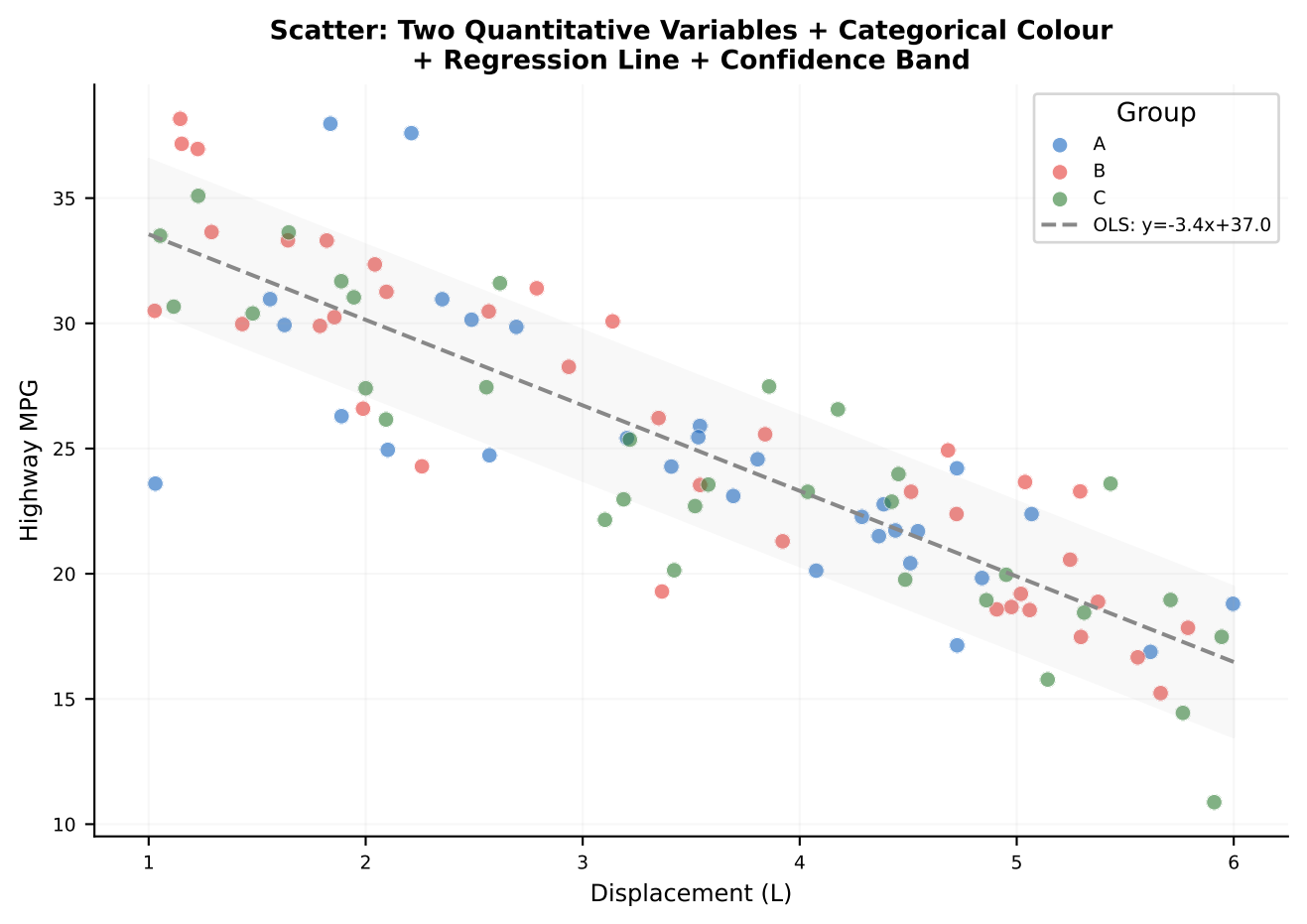
Module Roadmap

1. Scatter plot anatomy: position, colour, size, regression
2. Four correlation patterns and Anscombe's warning
3. Overplotting: alpha, jitter, hexbin, 2D density
4. Bubble charts: encoding a 3rd and 4th variable
5. Smoothers: linear, LOESS, and when to use which
6. Marginal distributions: scatter + histogram/rug

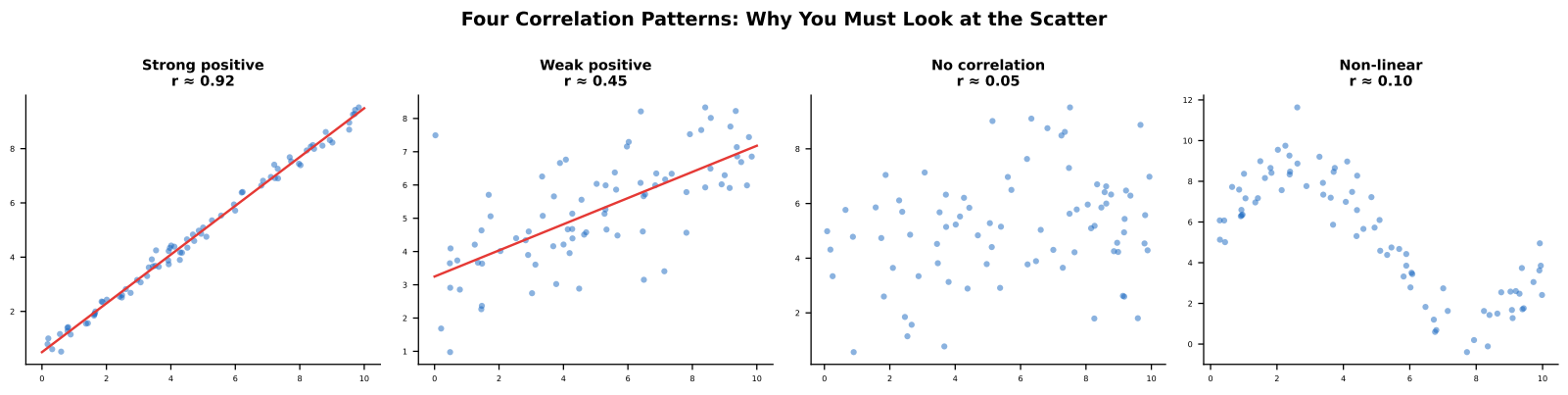
Learning Outcome
You will diagnose and fix overplotting, choose the right smoother, and produce Gapminder-style bubble charts encoding up to five variables.

Scatter Plot Anatomy

Position + Colour + Regression + Confidence Band



Four Correlation Patterns



Pro-Tip

Pearson's r misses non-linear relationships (panel 4). **Always plot the scatter** before trusting a correlation coefficient — this is Anscombe's lesson from W01-M01.

Scatter in R vs Python

R (ggplot2)

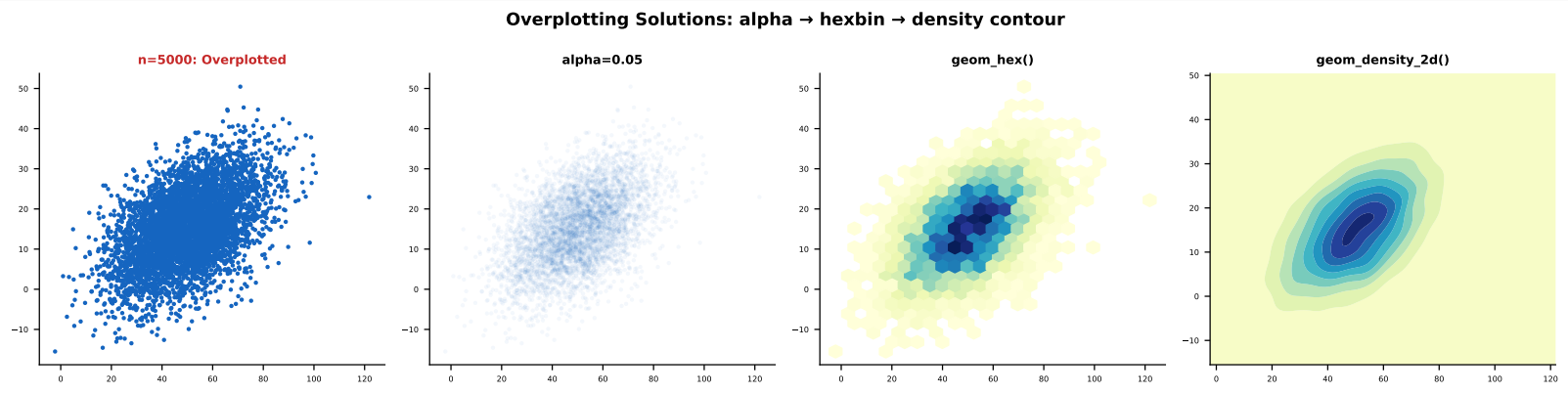
```r
ggplot(apps, aes(
    x = Reviews, y = Rating,
    color = Type)) +
  geom_point(alpha = 0.3,
    size = 0.8) +
  geom_smooth(method = "lm",
    se = TRUE, linewidth = 0.8) +
  scale_x_log10(
    labels = scales::comma) +
  scale_color_manual(values =
    c(Free = "#1565C0",
      Paid = "#E53935")) +
  theme_minimal() +
  labs(x = "Reviews (log)",
       y = "Rating")
```

Python

```python
# seaborn
sns.scatterplot(data=apps,
    x="Reviews", y="Rating",
    hue="Type", alpha=0.3, s=10,
    palette={"Free":"#1565C0",
             "Paid":"#E53935"})
ax.set_xscale("log")

# + regression
sns.lmplot(data=apps,
    x="Reviews", y="Rating",
    hue="Type", ci=95,
    height=5, aspect=1.3,
    scatter_kws={"alpha":0.3,"s"
        :8})

# matplotlib manual
ax.scatter(x, y, s=8, c=color,
    alpha=0.3, edgecolors="none")
m, b = np.polyfit(x, y, 1)
ax.plot(xs, m*xs+b, c="grey")
```

Overplotting

The Overplotting Cascade



Data Insight

As n grows: **alpha** works to $\sim$2K; **hexbin** works to $\sim$50K; **2D density contour** works for any n. The progression alpha $\rightarrow$ hexbin $\rightarrow$ density is a scale-adaptive overplotting solution.

Overplotting Solutions in Code

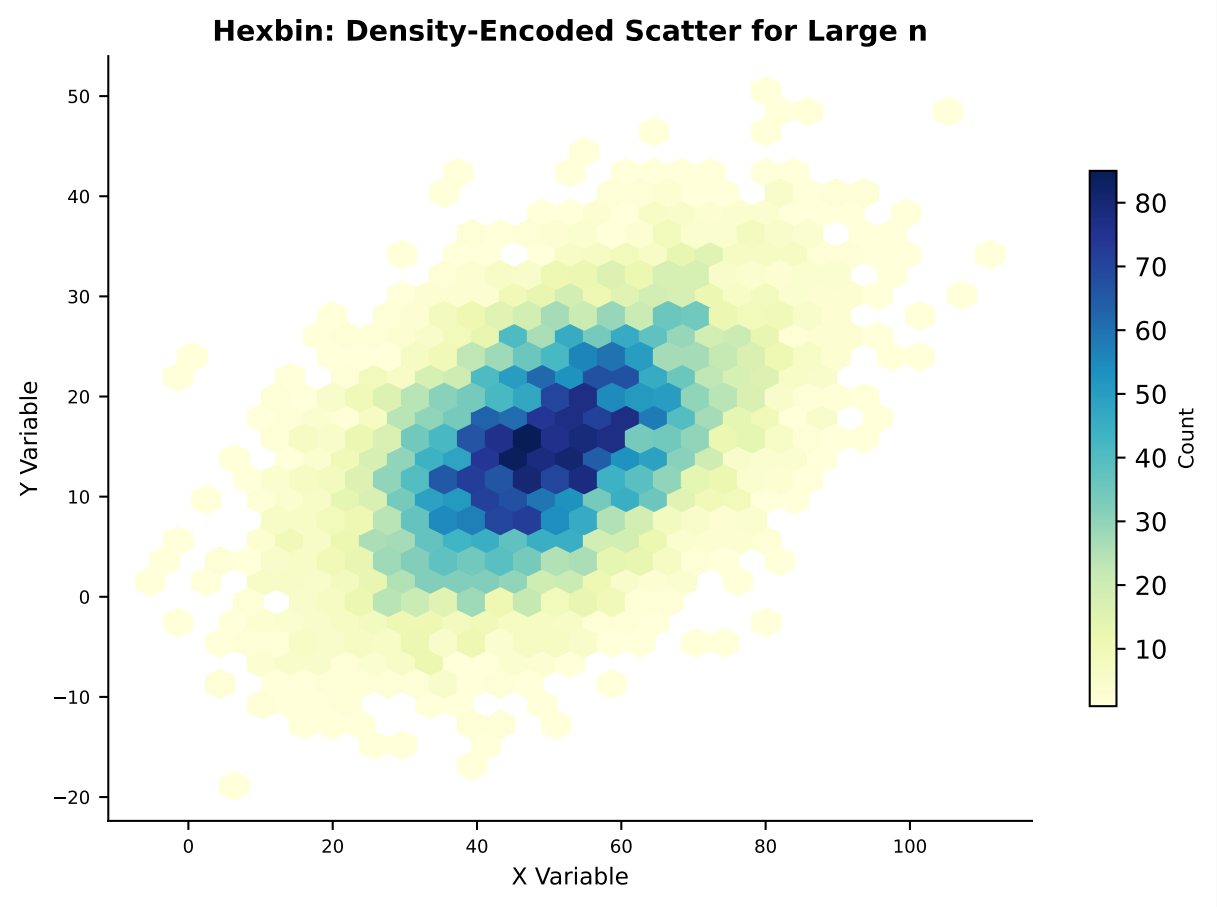
R

```r
# Alpha transparency
geom_point(alpha = 0.05, size =
    0.5)

# Hexbin
geom_hex(bins = 30) +
  scale_fill_viridis_c()

# 2D density contour
geom_density_2d(color = "grey50")

# 2D density filled
geom_density_2d_filled(alpha =
    0.7)
```

Python

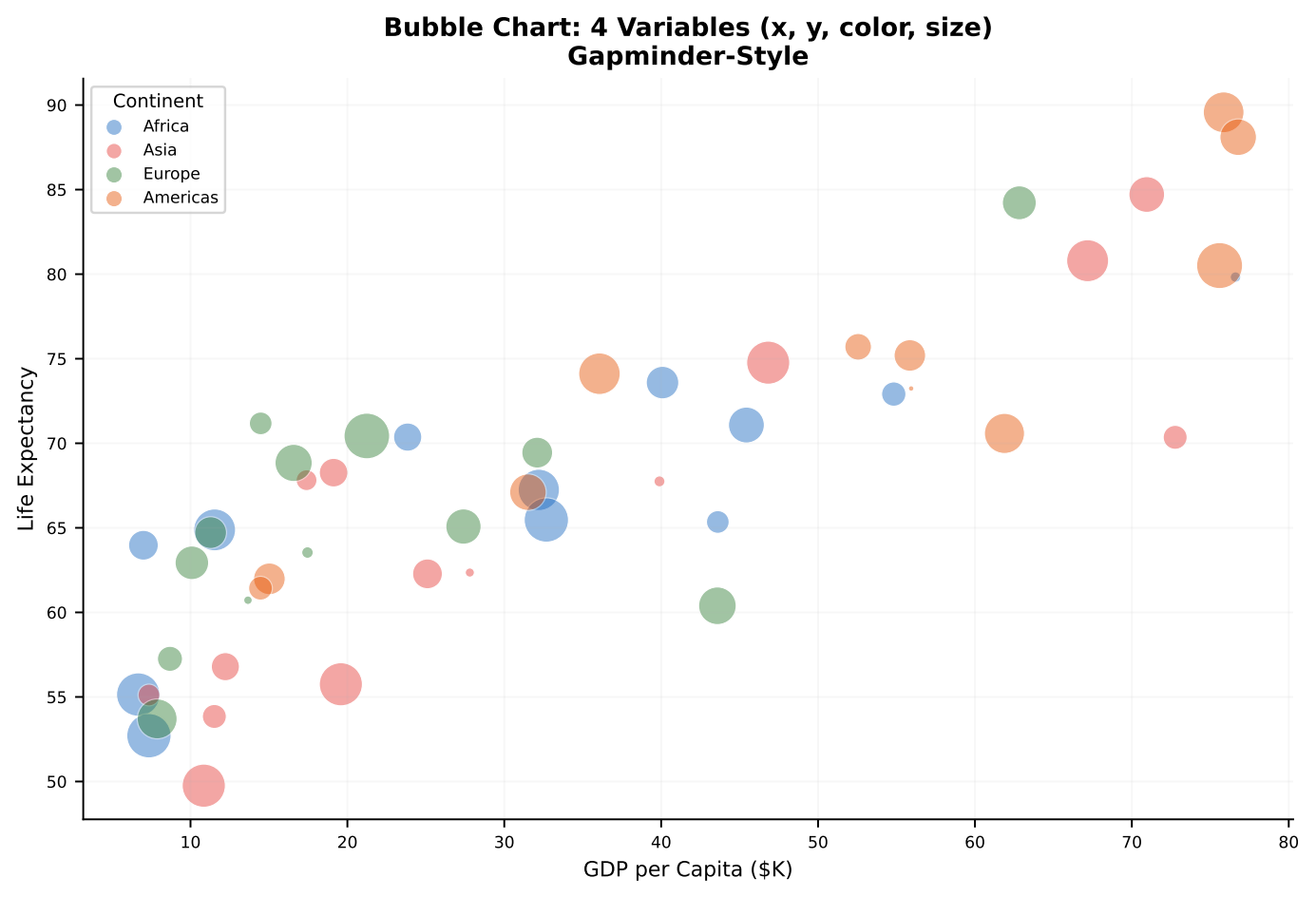
```python
# Alpha
ax.scatter(x, y, alpha=0.05, s=3)

# Hexbin
ax.hexbin(x, y, gridsize=30,
    cmap="YlGnBu", mincnt=1)

# 2D KDE contour
from scipy.stats import
    gaussian_kde
kde = gaussian_kde(np.vstack([x,y
    ]))
ax.contourf(xg, yg,
    kde(pts).reshape(xg.shape),
    levels=10, cmap="YlGnBu")

# seaborn jointplot
sns.jointplot(data=df, x="x", y="
    y",
    kind="hex")  # or "kde"
```

Hexbin: Density-Encoded Scatter



Bubble Charts

Bubble: 4 Variables in One Chart



Bubble Chart in Code

R

```r
ggplot(gap, aes(
    x = gdpPercap,
    y = lifeExp,
    color = continent,
    size = pop)) +
  geom_point(alpha = 0.5) +
  scale_x_log10(
    labels = scales::dollar) +
  scale_size(
    range = c(1, 15),
    guide = "none") +
  scale_color_brewer(
    palette = "Set2") +
  theme_minimal() +
  labs(title = "Gapminder",
    x = "GDP per Capita ($, log)"
        ,
    y = "Life Expectancy")
```

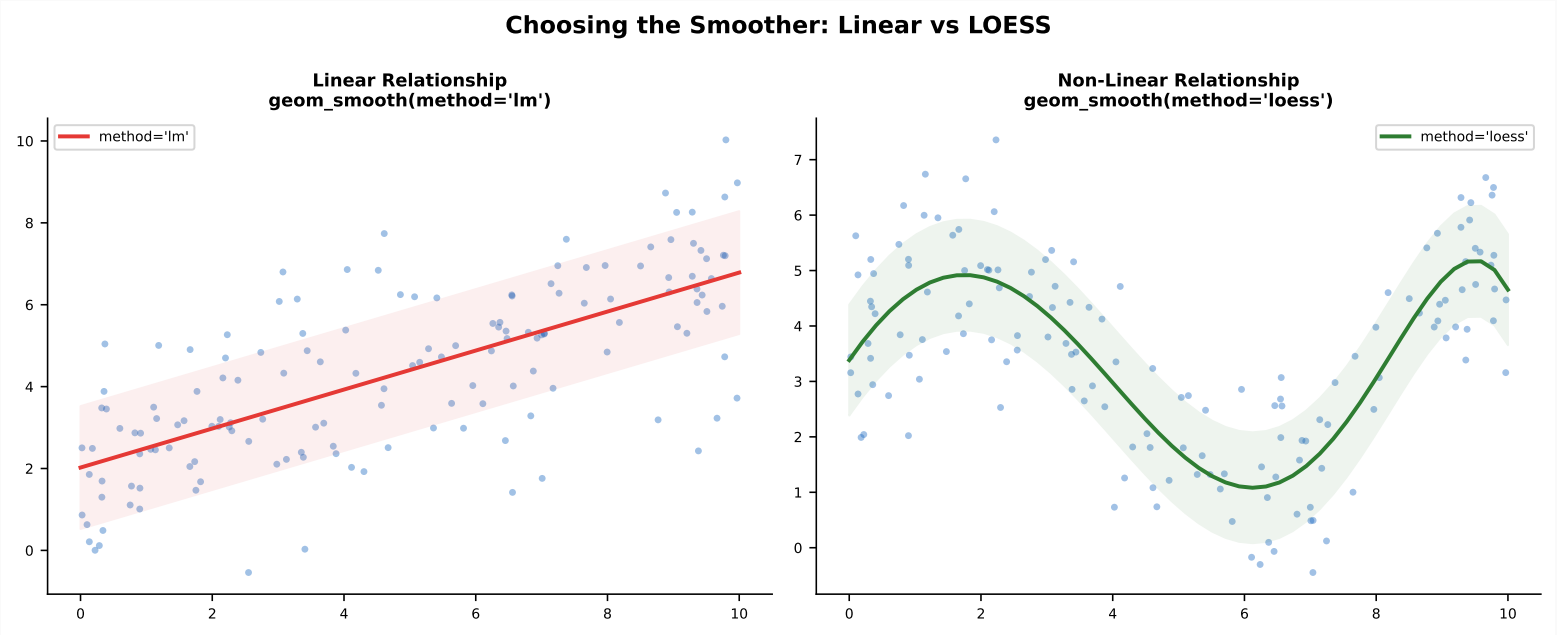
Python

```python
for cont, c in palette.items():
    m = df["continent"] == cont
    ax.scatter(
        df.loc[m, "gdp"],
        df.loc[m, "life_exp"],
        s=df.loc[m, "pop"] /
            scale,
        c=c, alpha=0.45,
        edgecolors="white", lw
            =0.5,
        label=cont)

ax.set_xscale("log")
ax.legend(fontsize=7,
    markerscale=0.5)

# Or plotly for interactive:
# px.scatter(df, x="gdp",
#     y="lifeExp", size="pop",
#     color="continent",
#     hover_name="country")
```

Pro-Tip
Map size to **area**, not radius: `scale_size()` in R and `sizemode/sizeref` in

Smoothers

Linear vs LOESS: Choosing the Right Smoother

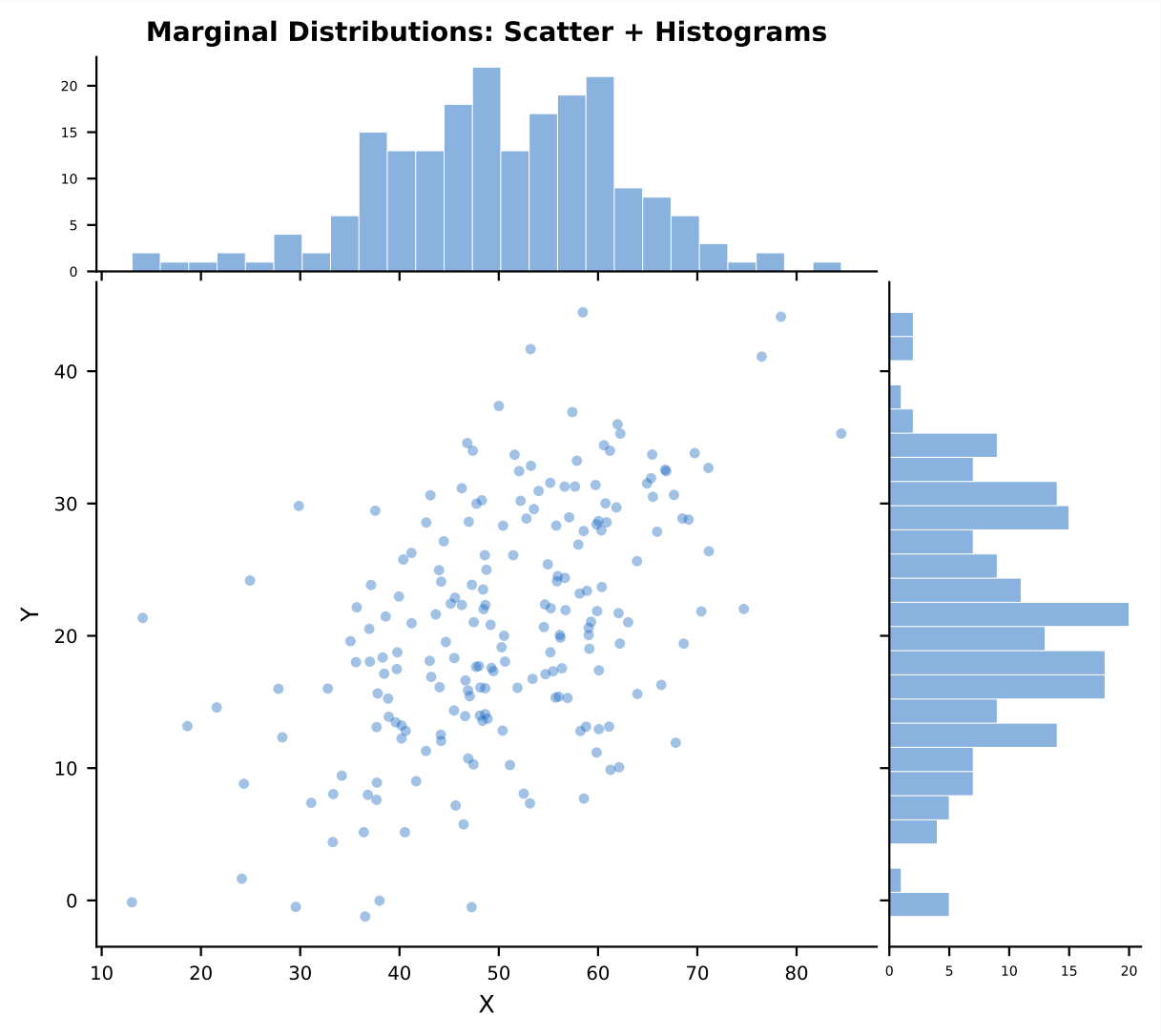


Data Insight

Use `method="lm"` when the relationship is approximately linear. Use `method="loess"` (default in ggplot2) or `method="gam"` for non-linear patterns. LOESS is local regression: it fits separate models in sliding windows, producing a flexible curve.

Marginal Distributions

Scatter + Marginal Histograms



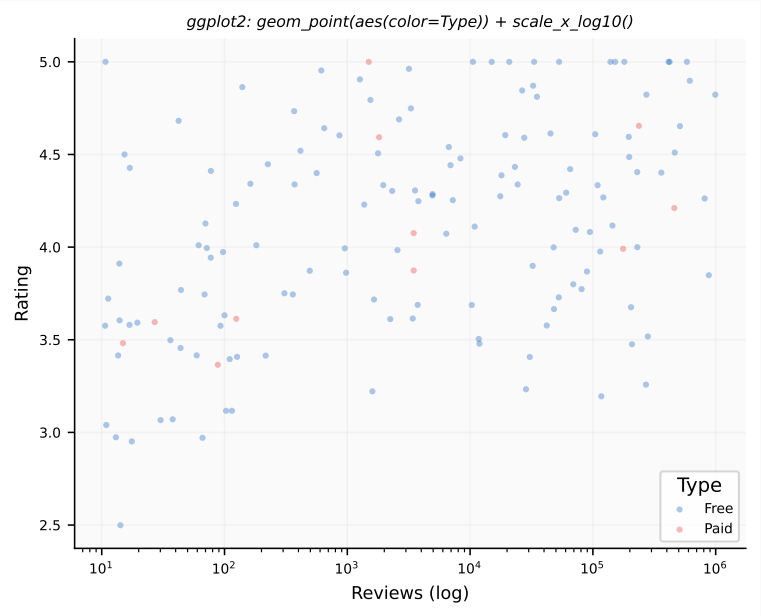
Pro-Tip

In R: `ggExtra::ggMarginal(p, type = "histogram")`. In Python:

Complete Examples

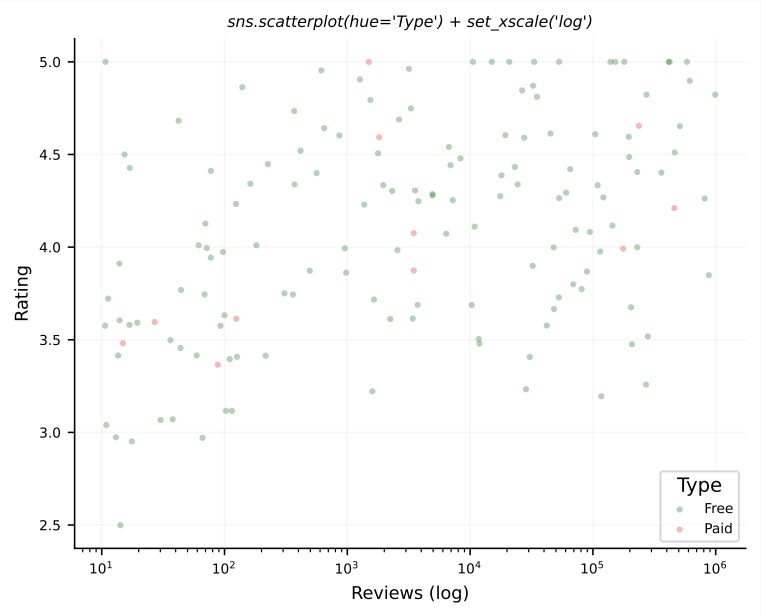
R: Reviews vs Rating (Publication Quality)

```r
apps_clean |>
  filter(!is.na(Rating),
         Reviews > 0) |>
  ggplot(aes(x = Reviews,
    y = Rating, color = Type)) +
  geom_point(alpha = 0.3,
    size = 0.8) +
  geom_smooth(method = "lm",
    se = TRUE, linewidth = 0.8) +
  scale_x_log10(
    labels = scales::comma) +
  scale_color_manual(values =
    c(Free = "#1565C0",
      Paid = "#E53935")) +
  coord_cartesian(
    ylim = c(1, 5)) +
  theme_minimal() +
  labs(
    title = "More Reviews
      = Higher Rating",
    subtitle = "Google Play Store",
    x = "Reviews (log)",
    y = "Rating",
    color = "Type")
```



Python: Same Chart

```python
fig, ax = plt.subplots(figsize
    =(6,5))

for t, c in [("Free","#2E7D32"),
             ("Paid","#E53935")]:
    m = apps["Type"] == t
    ax.scatter(
        apps.loc[m, "Reviews"],
        apps.loc[m, "Rating"],
        s=10, c=c, alpha=0.35,
        edgecolors="none", label=t)

ax.set_xscale("log")
ax.set_ylim(1, 5)
ax.legend(fontsize=7, title="Type")
ax.set_xlabel("Reviews (log)")
ax.set_ylabel("Rating")
ax.set_title(
    "More Reviews = Higher Rating",
    fontweight="bold")
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(
    False)
```



Synthesis

Relationship Chart Selection Guide

Relationship Chart Selection Guide

Situation	Chart	R	Python
2 continuous vars	**Scatter**	`geom_point()`	`ax.scatter()`
+ categorical group	**Colour scatter**	`aes(color=group)`	`c=palette[g]`
+ quantitative 3rd	**Bubble**	`aes(size=pop)`	`s=pop/scale`
Large n (>1K)	**Hexbin**	`geom_hex(bins=30)`	`ax.hexbin(gridsize=30)`
Very large n (>10K)	**2D density**	`geom_density_2d_filled()`	`ax.contourf(kde)`
Linear trend?	**+ lm smooth**	`geom_smooth(method='lm')`	`np.polyfit + ax.plot`
Non-linear trend?	**+ loess smooth**	`geom_smooth(method='loess')`	`LOWESS / polynomial`
Marginal dists?	**+ rug/hist**	`ggExtra::ggMarginal()`	`GridSpec + hist`

Key Takeaways

1. **Scatter** is the default for two continuous variables. Add colour for a 3rd (categorical), size for a 4th (continuous).
2. **Always plot the scatter** before trusting r — non-linear patterns and outliers are invisible in a number.
3. **Overplotting cascade**: alpha ($n < 2K$) $\rightarrow$ hexbin ($n < 50K$) $\rightarrow$ 2D density (any n).
4. **Bubble** encodes 4–5 variables; cap at 5 to avoid cognitive overload. Size maps to area, not radius.
5. **method="lm"** for linear; **method="loess"** for non-linear. LOESS is the ggplot2 default for $n < 1000$.
6. **Marginal distributions** add univariate context: `ggMarginal()` or `sns.jointplot()`.

Essential Readings

- Cleveland, W. S. & McGill, R. (1984). "Graphical Perception." *JASA*, 79(387), 531–554.
- Wickham, H. (2016). *ggplot2*, Chapter 5 (Scatter + Smoothers).
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapters 12–13 (Scatter, Bubble).
- Rosling, H. et al. (2018). *Factfulness*. (Gapminder bubble charts in context.)

Next Module: Proportions — Pie, Donut, Treemap & Waffle (W03-M05)